Collaborative Frequency-Aware Transformer for Unsupervised Multimodal Change Detection in Heterogeneous Remote Sensing Images

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***Abstract*—Multimodal change detection (MCD), as an emerging task, aims at recognizing change regions from bi-temporal remote sensing images (RSIs) of different modalities. Inspired by the success of the self-attention mechanism in the Transformer, attempts have been made to solve MCD through the Transformer variants. However, Transformer-based network optimization requires high-quality training samples. In addition, due to the significant differences in the data distribution, semantic information, and feature representation of multimodal data, Transformer-based methods have obvious deficiencies in local feature representation and spatial consistency, especially when dealing with heterogeneous images. To address the above challenges, we propose a collaborative frequency-aware Transformer for MCD (CFAT-MCD). As an unsupervised framework, CFAT-MCD is capable of learning more fine-grained patterns of land-cover change through a few pseudo-labels. The CFAT is designed to enhance spatial consistency and align the features on a multiscale (MS) basis, which can effectively mitigate the effects of modal differences. In addition, we propose a window-based spatial–frequency collaborative representation (SFCR) module to introduce frequency information into the spatial domain and improve the discriminability of spatial features. Extensive experiments on public datasets and quantitative analyses have validated the superior detection performance of our approach and the effectiveness of each module. The source code will be released at https://github.com/pu7yan9/CFAT_MCD.**

***Index Terms*—Multimodal change detection (MCD), Transformer, unsupervised learning.**

Received 19 February 2025; revised 16 June 2025 and 30 July 2025; accepted 21 August 2025. Date of publication 1 September 2025; date of current version 16 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62036006 and Grant 62376205, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2025JC-YBQN-800, and in part by the Scientific Research Program Funded by Shaanxi Provincial Education Department under Grant 24JK0350. *(Corresponding author: Maoguo Gong.)*

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Digital Object Identifier 10.1109/TGRS.2025.3604532

I. Introduction

With the rapid development of remote sensing (RS) technology and the increasing variety of sensors, change detection (CD) plays an important role in the fields of land-use monitoring [1], disaster assessment [2], environmental protection [3], and urban planning [4]. The basic single-modal CD method aims to extract feature change information from RS images (RSIs) acquired by the same sensor with the same modality. Due to the relative simplicity of the scene, the single-modal CD method has achieved remarkable results in the early research stage [5], [6], [7].

However, single-modal data are constrained by the imaging mechanism, which can provide information on only one physical property. Consequently, the single-modal approach exhibits limitations in complex change scenarios. To overcome this problem, researchers have begun to investigate multimodal data [multimodal CD (MCD)] methods, which enhance detection performance by integrating data from various sensors or modalities and leveraging the complementary attributes of each. For example, optical images provide rich textural and spectral information, while SAR images excel in capturing surface structures under all weather conditions. As an emerging task, MCD breaks the limitation imposed by the physical properties of the RSI, allowing CD techniques to operate effectively in more complex environments.

In recent years, various deep learning (DL) techniques have emerged, with the Transformer model being introduced to computer vision tasks owing to its exceptional performance in natural language processing [8], [9], [10], [11]. Additionally, the Transformer model has been extensively applied in RS tasks, attributed to its robust global modeling capabilities and proficiency in capturing long-range dependencies [12], [13], [14], [15]. Conventional DL-based CD methods rely on convolutional neural networks (CNNs) to extract local features [16], [17], [18], [19]. However, CNNs are constrained by a fixed receptive field, which limits their ability to capture global semantic information and spatial contextual relationships. CD typically involves surface changes in large-scale scenes, which requires models that can effectively capture both spatial and semantic global consistency. The Transformer model is regarded as a suitable solution to these challenges due to its capacity to flexibly model global features through

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the self-attention mechanism, and its inherent adaptability to multiscale (MS) variations.

However, research on Transformers in CD tasks has primarily concentrated on single-modal data [20], [21], [22], [23], [24], [25]. Several unresolved problems remain regarding effectively processing multimodal data using the Transformer model. Existing Transformer-based methods struggle to be applied in MCD and mainly face the following challenges.

1) High data requirement. The Transformer model requires high-quality training data to ensure the self-attention mechanism's effective operation, particularly in dense prediction tasks. The well-aligned labeled data are scarce in MCD.
2) Effect of modal differences. The global perception of the Transformer weakens its ability to capture local fine-grained features. In addition, modal differences will amplify the deficiencies in local features and spatial consistency of the model, thus affecting the detection performance.
3) Deficiencies of spatial feature. Existing models tend to deal with multimodal data by aligning and fusing spatial features. However, spatial features are sensitive to noise and unstructured variations, which makes spatial feature-based models susceptible to modal differences.

In previous work [26], [27], the introduction of frequency-domain information provides a novel perspective for addressing the limitations of spatial domain features. Unlike pixel-based or spatial-based approaches, frequency-domain representations enable the capture of structural patterns and textural characteristics in multimodal data from a global perspective, demonstrating particular proficiency in revealing subtle or heterogeneous changes across modalities. By decomposing multimodal data into spectral components, frequency-domain analysis enhances the discriminative capacity for periodic patterns (e.g., urban construction and lake expansion) while suppressing modality-specific noise, thereby improving the model's robustness against variations in imaging conditions.

The motivation of this study lies in designing a collaborative frequency–spatial-domain learning architecture that dynamically integrates frequency-domain information with spatial–semantic context, aiming to suppress noise interference while enhancing discriminative capacity for authentic changes. Specifically, our principal contributions can be summarized as follows.

1) In this article, we propose a collaborative frequency-aware Transformer for MCD (CFAT-MCD). As an unsupervised framework, CFAT-MCD is capable of learning more fine-grained patterns of land-cover change through a few pseudo-labels. Extensive experiments are carried out to validate the effectiveness of our approach.
2) Focusing on multimodal scenarios, our CFAT is designed to enhance the local feature and spatial consistency, thus improving the quality of CD maps. To fully utilize the advantages of the self-attention mechanism, CFAT extracts and aligns the features on an MS basis, effectively mitigating the effects of modal differences.
3) We also propose a spatial–frequency collaborative representation (SFCR) module to improve the discriminability of spatial features. The SFCR module maps the local frequency feature to the spatial domain using a window-based approach. Ablation experiments validate the impact of our SFCR module on the feature representation and detection performance.

The remainder of this work is constructed as follows. Section II briefly reviews the related work on MCD. Our proposed framework and the specific components are explicitly described in Section III. Sections IV and V report and analyze the comparison experiments and parameter analysis experiments, respectively. Conclusions and the future work are drawn in Section VI.

II. Related Work

A. Unsupervised Learning for MCD

The high cost and professional requirements associated with manual labeling pose significant limitations for supervised learning methods. In contrast, unsupervised learning enables applications in a wider range of scenarios by directly modeling unlabeled data and extracting potential patterns. Especially in heterogeneous scenes, unsupervised learning is able to capture correlations between different modalities. According to the type of core components, existing unsupervised approaches can be categorized into traditional, DL-based, and hybrid methods.

Traditional methods mainly rely on classical machine learning techniques such as mathematical modeling and statistical analysis to process images. For example, Mignotte [28] combines the geometric fractal decomposition and Markov random fields (MRFs) to capture the changes between multimodal data. The pixel difference maps of the bi-temporal images will be binarized in a Bayesian framework. Touati et al. [29] modeled the MCD task through the unsupervised Markovian framework. The main innovation is the proposed pixel pairwise-based observation field, which can measure the similarity between different modal images. Sun et al. [30] aligned the multimodal data in feature space through an iterative structural transformation to reduce the impact of modal differences on MCD. Then, the spatial context information is combined with pixel features using a conditional random field (CRF) model to generate the difference image (DI). Recently, they proposed a method that can directly output the change map without calculating the DI. The method maintains local consistency between superpixel pairs by a mathematical framework called the locally maintained energy model (LPEM) [31].

Meanwhile, DL methods mainly utilize network models such as the CNN, GNN, and GAN to learn useful features directly from data. Deep methods have stronger nonlinear modeling capabilities than traditional methods. Chen et al. [32] proposed an unsupervised method based on structural relational graph representation learning, which extracts reliable deep features through a graph structure learning method. The similarity of multimodal images is computed from the extracted graph representations to generate difference maps. Luppino et al. [33] proposed two deep image translation

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network frameworks that transform multimodal images into a unified representation space. The networks are then guided to learn the change patterns by an affinity-based change prior. Moreover, they proposed an unsupervised approach [34] based on code-aligned autoencoders, which captures and aligns pixel pair features by extracting domain-specific affinity matrices. Wu et al. [35] extracted common features between modalities by converting one modal image representation to another through the proposed commonality autoencoder network. The common feature is used to generate the DI and enable change segmentation. Han et al. [36] proposed a structure difference compensation graph autoencoder (SDC-GAE) to learn the structural differences between multimodal images. The compensation value directly reflects the intensity of land-cover changes. The composite loss, consisting of three components, guarantees the consistency of the structural features. Recently, Liu et al. [37] proposed a novel superpixel-based autoencoder Kolmogorov–Arnold network (AEKAN). The AEKAN explores common features in unchanged regions through the KAN and quantifies the change magnitude through the mean square error (MSE).

Besides, there are hybrid methods, which utilize the idea of multimodal, multitasking to synergize traditional and deep methods to improve detection performance or computational efficiency. Jiang et al. [38] proposed a novel homogeneous transform model based on image style migration (IST), which ensures that the bi-temporal features can complement each other by transforming the semantics and style of the heterogeneous images through traditional image algorithms and a convolutional network, respectively. The land-cover change is then detected by using the fused features. Radoi and Ünsalan [39] constructed a location dictionary of self-similar structures using a modified fractal coding technique. Then, the feature extraction and compression of the self-similar structures is performed by CNNs. The features are used to transform the image modality and generate the change map by the unsupervised expectation maximization (EM) method.

In contrast to the aforementioned approaches, our CFAT-MCD utilizes the pseudo-labels generated by traditional detection methods as a priori knowledge. It further enhances the performance of traditional detection methods by leveraging the learning of collaborative space–frequency features through a deep Transformer network.

B. Frequency-Domain Learning in the RSI

Frequency-domain information describes the intensity and distribution of different frequency components in an image. Unlike spatial features, frequency-domain features can directly reflect the pattern of change, revealing information about the image's structure, texture, and detail. Frequency-domain learning provides new solutions for various computer vision tasks. For example, Patro et al. [40] proposed a novel architecture called SpectFormer, which combines spectral and multihead attention layers based on Vision Transformers (ViTs). Introducing a frequency-domain feature extraction module and an attention fusion mechanism in the Transformer improves its ability to capture local details and global context information. Finder et al. [41] proposed an approach combining wavelet transform and convolutional operations to separate global and local features by wavelet decomposition. This method improves the model's global modeling ability and fine prediction performance in complex tasks while reducing the computational complexity.

In the RSI, frequency-domain features have a natural advantage in dealing with noise and interference from modal differences. Consequently, an increasing number of researchers have realized the improvement of the accuracy and robustness of RS tasks by utilizing frequency-domain information. Zheng et al. [26] designed a high-frequency attention module and combined it with a Siamese network structure for contrastive learning of change regions. The introduction of frequency-domain information enhances the sensitivity of the model to small changes and reduces the noise interference, thereby improving the accuracy and robustness of the detection network. Zhou et al. [27] compensated for the loss of image details caused by downsampling by introducing the frequency information. They proposed a joint frequency–spatial domain network (JFSDNet) to reduce the complexity of frequency features through a frequency selection module that discriminates and selects frequency information adaptively. Xue et al. [42] addressed the difficulty of Transformer-based methods in capturing the long-range dependency of global features through a triple CD network. In this network, a multifrequency channel attention module is proposed to enhance the ability to model channel correlation, which can compensate for the underrepresentation of features caused by performing only global average pooling. Xu et al. [43] revisited the CD task from a signal-processing perspective, defining it as a coherent detection task of two 2-D signal distribution features. The authors construct a dual-frequency feature encoder Siamese framework, which uses a CNN and an attention mechanism to capture local high-frequency signals and global low-frequency signals after splitting the input RS image into two channels. The method combines a multiscale integrated network to fuse different frequency features with multiscale spatial features to describe the changing region comprehensively. Guo et al. [44] extracted frequency difference features at different scales and fuse spatial- and frequency-domain features through the channel attention mechanism, which enhances the network's ability to detect change edges and insignificant regions.

Inspired by frequency-domain learning, we propose a dense approach to combine spatial and frequency features. The collaborative module adaptively filters valuable information in the frequency domain and maps it to the spatial domain via a window-based operation. The co-representation features are effectively enhanced in both modality match and change discriminability.

III. Methodology

In this section, we will introduce the overall structure of the proposed unsupervised MCD method and describe each component in detail.

A. Overall Structure

The CFAT-MCD, as an unsupervised framework, is able to achieve more fine-grained detection of feature changes

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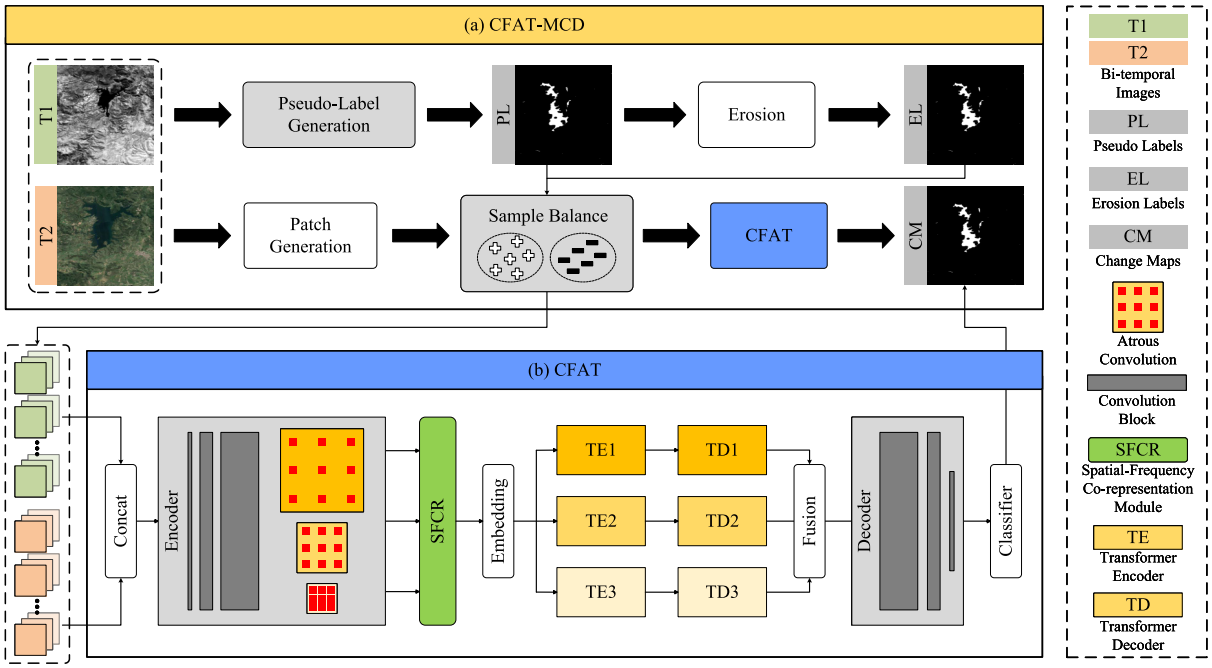


Fig. 1. Overall framework of our CFAT-MCD. We present the complete workflow of CFAT-MCD and its components in (a). The network structure of the CFAT is shown in (b).

with a small amount of pseudo-labeling information. The overall framework of CFAT-MCD is shown in Fig. 1(a), which contains three main parts: pseudo-label generation, training sample generation, and detection network. The pseudo-label generation part can be regarded as a coarse CD process based on traditional methods. The results of the coarse detection are used as pseudo-labels for CFAT training, which is a regular approach in unsupervised learning. However, a notable limitation of this approach is that the quality of the pseudo-labels may constrain the training performance of the detection network. Pseudo-labels with low confidence can mislead the network's learning, particularly during the early stages of training. To solve this problem, we design the erosion and sample balance modules for the training sample generation part. Label erosion achieves denoise and boundary smoothness by performing local processing on each pixel. The eroded labels have higher internal consistency, which can improve the completeness of the change region. However, there are still a number of low-confidence samples with missed or false detection in the pseudo-label. In the sample balance module, we mark patches as positive and negative by comparing the pseudo- and erosion labels. We filter out positive samples and control the number of negative samples to be consistent with positive samples to achieve a balanced training sample. The balanced pseudo-labels with the corresponding patches are used to train our CFAT model. The CFAT is a patch-to-pixel detection network proposed to generate each patch's change intensity (CI). The predictions of all patch samples form the complete CI map (CIM). The CIM is then transformed into a binary CD image (BCDI) by the threshold segmentation operation, which is the final output of CFAT-MCD.

Specifically, the bi-temporal images $T1$ and $T2$ are used to generate the pseudo-label Y_p of the land-cover change by the traditional method P_{tra}. Any existing work, such as [28], [29], [30], [31], [45], [46], [47], [48], and so on, can be used as P_{tra} to compose our CFAT-MCD. These traditional methods show differences in detection performance, resulting in varying quality of pseudo-labels. We performed a series of processes on Y_p in the training sample generation part, which can suppress the impact of quality differences on the detection network. In the training sample generation part, we use the structure element S to perform the erosion operation on the binary label Y_p, which is computed as follows:

$$Y'_p = \left(Y_p \ominus S\right)(z) = 1 \quad \text{if} \quad S_z \subseteq I, \quad \text{else} \quad 0 \tag{1}$$

where S_z is a shifted version of the structure element S with center at position z. The output value is one when all the foreground pixels of the structure element S match the foreground pixels at the corresponding positions in the label Y_p. If any of them do not match, the output is zero. The erosion operation aims to reduce false detections in pseudo-labels by compressing the boundaries of the changing regions and eliminating small-scale discrete changes. Patch generation is the process of cropping the bi-temporal images into patches pixel by pixel. In the sample balance module, we mark the patches by comparing the values of Y'_p and Y_p at the same positions. If the values of the erosion label and pseudo-label are the same change classes, this patch will be marked as a positive sample. Otherwise, mark the patch as a negative sample. Finally, we randomly select the same number of negative samples based on the number of positive samples to form the complete training set. The CFAT is a hybrid network that combines a convolutional network and a Transformer,

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TABLE I
DETAILED DESCRIPTION OF THE CFAT MODEL

	Stage	Stage output	Layers
Encoder	1	$64 \times H_p \times W_p$	$Conv(6, 64), BN, ReLU$
	2	$256 \times H_p \times W_p$	$Conv(64, 256), BN, ReLU$
	3-1	$64 \times H_p \times W_p$	$Conv(64, 256, 2, 2), BN, ReLU$
	3-2	$64 \times H_p \times W_p$	$Conv(64, 256, 4, 4), BN, ReLU$
	3-3	$64 \times H_p \times W_p$	$Conv(64, 256, 6, 6), BN, ReLU$
SFCR	4	$256 \times H_p \times W_p$	$SFCR(64, 256, (3, 3))$
Transformer	5-1	$L_t \times dim$	$TransEn(256, 1, 4, 256, 256)$
		$256 \times H_p \times W_p$	$TransDe(256, 1, 4, 256, 256)$
	5-2	$L_t \times dim$	$TransEn(256, 1, 4, 256, 256)$
		$256 \times H_p \times W_p$	$TransDe(256, 1, 4, 256, 256)$
	5-3	$L_t \times dim$	$TransEn(256, 1, 4, 256, 256)$
		$256 \times H_p \times W_p$	$TransDe(256, 1, 4, 256, 256)$
Decoder	6	$256 \times H_p \times W_p$	$Conv(768, 256), BN, ReLU$
	7	$64 \times \frac{H_p}{2} \times \frac{W_p}{2}$	$Conv(256, 64, 4, 4), BN, ReLU$
Classifier	8	$64 \times 1 \times 1$	$AvgPool(1)$
	9	1	$Linear(64, 1)$

which can learn the change patterns of multimodal images from the pseudo-labeling training process, thereby generating more refined BCDIs.

B. Collaborative Frequency-Aware Transformer

For multimodal scenarios, our proposed CFAT combines the ideas of hybrid models and MS to make up for the deficiencies of the Transformer model in terms of local features and spatial consistency. The CFAT consists of three independent Transformer networks and a pair of encoder–decoder networks. We present the complete details of the CFAT components in Table I. The encoder network processes the input patches and extracts the low-level features, the Transformer network then performs global perception of the features, and finally, the decoder network uses the extracted local–global features for CI prediction. The overall framework of the CFAT is shown in Fig. 1(b), which is a patch-to-pixel detection model whose inputs are bi-temporal patches $X_1, X_2 \in \mathbb{R}^{C_p \times H_p \times W_p}$, and the outputs are the classification results of central pixel in each patch. First, X_1 and X_2 are fed into the encoder network to extract the shallow features after concatenating. The encoder network consists of two base convolutional layers and one MS convolutional layer. The patch size is kept constant during the convolution process, and the feature channels are expanded according to 64-256-64. We follow the setting of ASPP [49] in the MS convolutional layer, where feature extraction is performed in parallel by three convolutional layers with different dilation rates. The purpose of this design is to capture features under varying receptive fields while preserving local spatial detail in the input patches. For dense prediction tasks such as CD, it is crucial to integrate both local fine-grained and global contextual information to improve discriminative feature representation. To this end, we design an MS perception structure using dilated convolutions with varying dilation rates. Given a shallow input feature map $F_{\text{en}} \in \mathbb{R}^{C \times H \times W}$, we apply three parallel 2-D convolutions with a kernel size $k = 3$ and dilation rate $r_s \in \{2, 4, 6\}$. The computation of the MS feature map $f_s(i, j)$ at the spatial location (i, j) under the dilation rate r_s can be formulated as

$$f_s(i, j) = \sum_{m}^{k-1} \sum_{n}^{k-1} W_{\text{s}}(m, n) \cdot F_{\text{en}}(i + r_{\text{s}} \cdot m, r_s \cdot n) \quad (2)$$

where W_s represents the convolution kernel weights corresponding to the dilation rate r_s. To preserve the spatial dimensions of the output feature maps and enable element-wise fusion, we set the padding size for each convolutional branch same as r_s. The use of dilated convolution expands the effective receptive field without increasing the number of parameters or losing resolution, enabling the network to aggregate context from a broader area. Features extracted from different scales will be concatenated after the Transformer operations and sent to the decoder network to obtain a unified representation. The feature F_s at each scale will be collaboratively represented with the frequency-domain information in the SFCR module to obtain F_{co}, and we will introduce the computation process of SFCR in Section III-C. F_{co} is flatten and embedded into the token through the embedding operation

$$X_t = A' \cdot F_{\text{co}}^T = \text{Softmax}\left(\text{reshape}\left(\text{Conv}\left(F_{\text{co}}\right)\right)\right) \cdot \text{reshape}\left(F_{\text{co}}\right). \quad (3)$$

For the co-representation feature F_{co}, we compute the spatial attention weight A of F_{co} by a convolution operation. Then, A is flatten and normalized to get A' of size $(L_t, H \times W)$. We keep the channel dimension of F_{co} and flatten the spatial dimension to obtain F'_{co} of size $(C, H \times W)$. Finally, the token representation applicable to the Transformer is obtained by computing the product of A' and F'_{co} transpose, when the token size is (L_t, C). In contrast to the standard 1-D positional embeddings used in the ViT [10], we employ a multilevel, independent positional embedding strategy in conjunction with an MS structural design. Both embedding methods are nonfixed approaches based on learnable parameters. The difference lies in the location and shared scope. The position encoding in the ViT is globally shared, whereas we perform independent position encoding at each level, allowing us to obtain spatial position information at different granularities. This design is particularly well-suited for MCD tasks where the modality gap exists. The learnable positional encodings are adaptively updated through gradient-based optimization, thereby providing more stable and informative spatial representations. In the Transformer part, we learn the co-representation feature under different receptive fields by three independent networks. The aim is to model the global features of changing targets at different scales by taking advantage of the self-attention mechanism. We follow the structure of the classical Transformer; the details of each component are shown in Fig. 2. Each Transformer network consists of an encoder block and a decoder block.

1) Transformer Encoder: As the core module for processing token sequences and extracting global features, the module consists of L stacked encoder blocks. In our approach, we use the structure of a single encoder block, and each block mainly contains the LN, MHSA, and FFN. The embedded token X_t is fed into the MHSA module after LN processing

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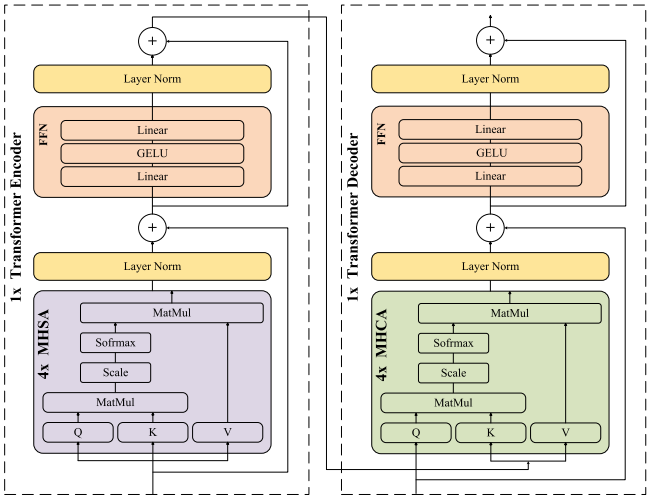


Fig. 2. Framework of our Transformer block. It follows the standard structure of the ViT but removes the dropout operation.

to learn the global dependencies of the sequence. MHSA is based on the self-attention mechanism, which captures the region correlations within a patch by calculating the similarity between different positions in the token. In MHSA, each head is a separate scaled dot-product attention operation, which is transformed by a linear transformation, dot-product attention, scaling, and weighted summation to get the final output. First, the query Q, key K, and value V are computed by three sets of linear transformations

$$Q = X_t W_Q,\ K = X_t W_K,\ V = X_t W_V \tag{4}$$

where W_Q, W_K, and W_V are the learnable parameter matrices of size (C, d_k). d_k is the dimension of each head, which we set to $d_k = (C/N_{\text{head}}) = 64$. The similarity scores are then computed through Q and K as

$$\text{scale_socre} = \frac{QK^T}{\sqrt{d_k}}. \tag{5}$$

This score indicates the similarity of each position in the patch to the other positions. $(1/(d_k)^{1/2})$ is used as a scaling factor to prevent the value of the dot product from being too large, and this scaling operation benefits the steady flow of the gradient. The scaled dot product result is converted to a probability distribution by the Softmax function to obtain the attention weight W_{att}. Finally, the output of one head is obtained by a weighted summation of the W_{att} with the V

$$\text{Attention}(Q, K, V) = W_{\text{att}} \cdot V = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V. \tag{6}$$

For each head in the MHSA, the above computational process is performed independently so that multiple heads can capture different attention information. The final outputs of MHSA are computed as follows:

$$\text{MHSA}(X_t) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_N) W_o. \tag{7}$$

In this article, we set the number of heads N to 4. The output of the MHSA is connected to the original input by means of residuals after LN processing

$$\tilde{X}_t = X_t + \text{LN}(\text{MHSA}(X_t)). \tag{8}$$

X_t is then fed into the FFN for feature enhancement and nonlinear representation. The FFN improves the expressive ability of X_t through two linear layers and a nonlinear activation function, GELU, thus helping the model to fit the complex data better. The computation process is as follows:

$$\text{FFN}(\tilde{X}_t) = \sigma(\tilde{X}_t W_1 + b_1) W_2 + b_2. \tag{9}$$

Similarly, the final output of the Transformer encoder is obtained after LN and residual concatenation

$$\text{Output}_{\text{en}} = \tilde{X}_t + \text{FFN}(\tilde{X}_t). \tag{10}$$

It is worth noting that we do not use the dropout regularization operation in the Transformer. This is because randomly setting the outputs of part neural nodes to zero will cause the loss of spatial information, thereby disturbing the consistency of spatial features. For tasks like MCD that require fine-grained feature perception, the dropout operation reduces the granularity sensitivity of the model.

2) Transformer Decoder: In the decoder section, the network components remain the same as the encoder, except that the MHSA module is replaced with the multihead cross-attention (MHCA) module. The computation process of MHCA is consistent with (6). The difference is that Q is from the original input X_t, while K and V are from the output of Transformer encoder

$$Q = X_t W_Q,\ K = \text{Output}_{\text{en}} W_K,\ V = \text{Output}_{\text{en}} W_V. \tag{11}$$

The outputs of MHCA and FFN are also connected in a residual way after passing through the LN operation. For the features of three different receptive fields $f_s^i \in \mathbb{R},\ i = 1, 2, 3$, we model the global information through three identical and independent Transformer networks to get $f_{\text{tra}}^i \in \mathbb{R},\ i = 1, 2, 3$. We concatenate the three sets of features and send them into the decoder network for task prediction. In the decoder network, we perform a deeper semantic mapping of the MS features through two convolutional layers. The spatial dimension is then compressed by an adaptive average pooling operation. Eventually, the CI prediction P is realized by a linear classifier.

3) Loss Function: The network parameters of our CFAT are optimized by the standard binary cross-entropy (BCE) loss. The BCE loss is formulated as follows:

$$\begin{aligned}\mathcal{L}_{cd} &= \text{BCE}(Y'_p, P) \\ &= -\frac{1}{N}\sum_{i=1}^{N}\left[y_p^i \cdot \log(p^i) + (1 - y_p^i) \cdot \log(1 - p^i)\right]\end{aligned} \tag{12}$$

where Y'_p is the erosion pseudo-label and N is the total number of training samples. We summarize the training process of the CFAT and show it in Algorithm 1.

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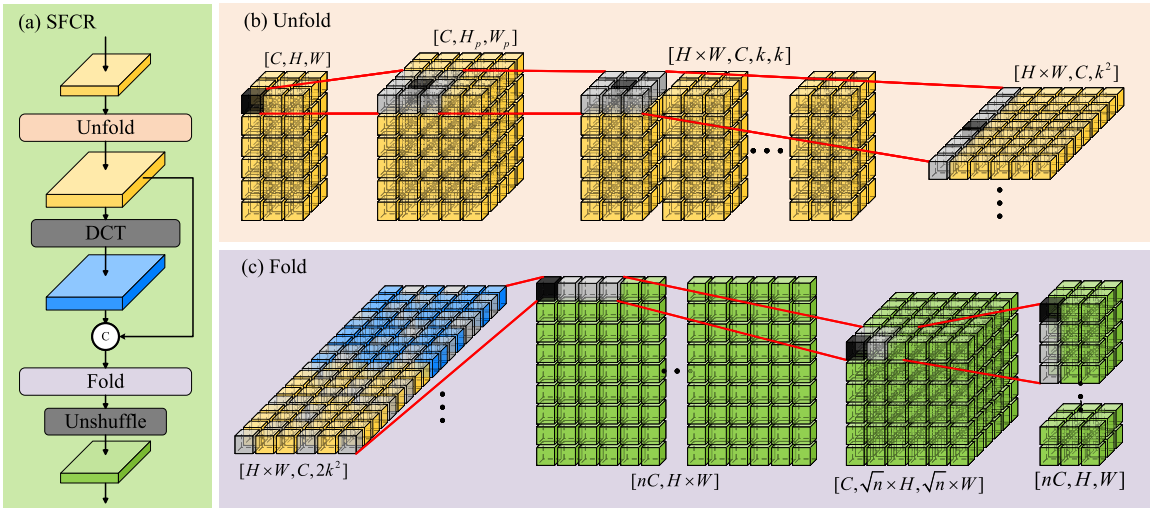


Fig. 3. Structure of our SFCR module. (a) SFCR is a pixel-level collaborative representation method that maps frequency-domain features into spatial features in a window-based manner. The process details of two core operations are presented in (b) and (c).

Algorithm 1 Training Process of the CFAT

Input: P_1, P_2: The balanced samples of bi-temporal patches with different modality, Y: the balanced samples of pseudo label, E_c, D_c: the encoder and decoder of CNN, E^i_{tra}, D^i_{tra}: the i sets of Transformer encoder and transformer decoder, C: the classifier.

for *epoch* to *Epoch* **do**
  Concatenate the P_1 and P_2.
  Extract shallow feature F_{en} through encoder network E_c.
  Extract MS spatial feature f^i_s by Eq.(2).
  for i to *Scale*=3 **do**
    Extract frequency feature f^i_{fre} by Eq. (14).
    Generate frequency-aware feature f^i_{co} by Eq. (17).
    Feature embedding to generate token x^i_t by Eq. (3).
    Capture global correlation through the E^i_{tra} and D^i_{tra}.
  end for
  Fuse the f^i_{tra} from different scales.
  Compress the F_{tra} through D_c.
  Generate the CIM through classifier C.
  Calculate CD loss $\mathcal{L}_{cd}$ with Y by Eq. (12).
  Propagate back the $\mathcal{L}_{cd}$ and update optimizer.
end for

Output: $\mathcal{L}_{cd}$: The CD loss, $\widehat{\theta}_e$, $\widehat{\theta}_d$: parameters for E_c, D_c, $\widehat{\theta}^i_{te}$, $\widehat{\theta}^i_{td}$: parameters for E^i_{tra}, D^i_{tra}

C. Spatial–Frequency Co-Representation Module

Inspired by frequency-domain learning, we propose a novel SFCR module that effectively improves the comparability of multimodal features by introducing frequency-domain information to the spatial domain. Existing spatial–frequency methods [26], [27], [42], [43] tend to introduce frequency information in the feature channel dimension and then fuse the features by different attention strategies. Unlike these methods, our SFCR is a pixel-level collaborative representation method with high fine-grained granularity, which maps frequency-domain features into spatial features in a window-based manner. This collaborative representation can be seen as a deeper form of feature fusion, and its operation process is shown in Fig. 3. Modal differences lead to mismatches between feature representations and feature types in the spatial domain. Therefore, it is difficult to generate BCDIs directly by making difference comparisons. While in the frequency domain, the feature representation describes the spatial distribution pattern and change characteristics of the features through the spectrum frequency. Land-cover changes will affect the energy distribution of the image spectrum, and such disturbances are not sensitive to modal differences. Thus, frequency-domain features possess more robust comparability than spatial features in multimodal scenes. With appropriate fusion strategies, collaboratively modeling spatial- and frequency-domain features allows the network to better capture texture changes and structural differences between modalities or temporal observations. For the MS spatial feature map $F_s \in \mathbb{R}^{C\times H\times W}$, the SFCR module proceeds through three stages: spatial unfolding, local discrete cosine transform (DCT), feature fusion, and reconstruction. We first apply a sliding window of size $k \times k$ to extract overlapping spatial patches from the padded feature map. This operation transforms each patch into a column vector of size Ck^2, resulting in an unfolded representation

$$F_{un} = \text{Unfold}\left(\text{Pad}\left(F_s\right)\right) \in \mathbb{R}^{Ck^2\times HW}. \tag{13}$$

Each column corresponds to a local patch vector centered at a spatial position, preserving local spatial dependencies in a vectorized form. For each local patch $f_{un} \in \mathbb{R}^{k\times k}$ from a single channel, we compute its frequency-domain representation using the 2-D DCT. This operation captures localized frequency information such as edge directionality and texture granularity. The transformation is defined as

$$f_{\text{fre}}(u,v) = \sum_{i=0}^{k-1}\sum_{j=0}^{k-1} f_{un}(i,j)\cdot G_{\text{kernel}}(u,i)\cdot G_{\text{kernel}}(v,j) \tag{14}$$

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TABLE II
DETAILED DESCRIPTION OF THE FOUR HETEROGENEOUS CD DATASETS ADOPTED IN THE EXPERIMENTS

Datasets	Date	Modality	Size (used band)	Spatial resolution (m)	Change Event	Location
#1	Sept. 1995/Jul. 1996	Landsat-5/Google Earth	$300 \times 412 \times 1/3$ (all)	30	Lake Expansion	Sardinia, Italy
#2	Jun. 2008/Sep. 2010	Radarsat-2/Google Earth	$343 \times 291 \times 1/1$ (all)	8	Embankment	Yellow River, China
#3	Jun. 2008/Sep. 2012	Radarsat-2/Goole Earth	$593 \times 921 \times 1/3$ (all)	8	Constructions	Shuguang, China
#4	1999/2000	Spot/NDVI	$990 \times 554 \times 3/1$ (all)	≈ 25	Flooding	Gloucester, UK
#5	May 2012/Jul. 2013	Pleiades/WorldView2	$2000 \times 2000 \times 3/3$ (all)	0.52	Construction	Toulouse, France

where $G_{\text{kernel}} \in \mathbb{R}^{k\times k}$ is the orthogonal DCT kernel matrix, used to perform separable 2-D DCT via matrix multiplication. Each element of the kernel matrix is given by

$$G_{\text{kernel}}[u,x] = C(u)\cdot\cos\left(\frac{\pi}{2k}\cdot u\cdot(2x+1)\right),\ 0\le u,\ x<k. \tag{15}$$

Here, x represents the position index in the window and u is the transpose of x, representing the frequency index in the kernel function. The normalization matrix C_u ensures orthogonality and energy consistency of the DCT, which is computed as follows:

$$C(u) = \sqrt{\frac{\max(2-u,1)}{k}}. \tag{16}$$

This DCT operation is applied independently to each sliding window in all channels, producing frequency features F_{fre}, aligned with the unfolded spatial feature F_{un}. The spatial feature and the frequency feature are concatenated along the channel dimension and passed through a learnable linear projection layer to achieve dimensionality reduction and integration. Finally, the fused vectorized features are folded back into the spatial layout using the inverse of the unfolding process

$$F_{\text{co}} = \text{Fold}(\text{Fuse}(F_{un}, F_{\text{fre}})) \in \mathbb{R}^{C\times H\times W}. \tag{17}$$

Here, the fold operation reconstructs the spatial feature map by aggregating overlapping windows. The channel of feature F_{co} is mapped back to size C through a linear layer. Through the calculations performed by the SCFR module described above, spatial features and frequency features can complement each other more effectively. The extracted features enhance the details of land-cover changes while maintaining spatial consistency.

IV. EXPERIMENTS AND RESULTS

In this section, we conduct a series of experiments to test the performance of the proposed approach. This section specifically includes the following contents. First, we provide an introduction to the experimental dataset and evaluation metrics. Second, eleven benchmark methods are used for comparison, and the experimental settings are described. Then, the CIMs and BCDIs of different approaches are compared to verify the superiority of our proposal. Finally, we verify the robustness and effectiveness of CFAT-MCD based on the initial results of different methods.

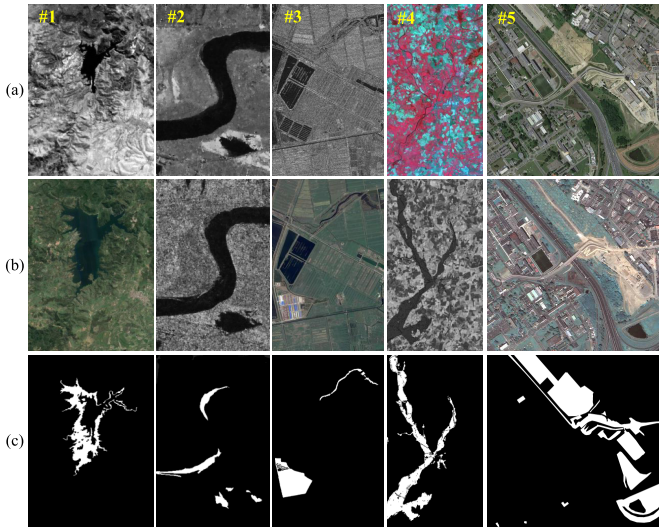


Fig. 4. Experimental datasets display. (a) t_1-time image, (b) t_2-time image, and (c) ground-truth image. #1–#5 represent the four MCD datasets corresponding to those in Table II.

A. Dataset Descriptions and Evaluation Indicators

1) Dataset Descriptions: In our experiments, we validate the effectiveness of our CFAT-MCD with five widely used MCD datasets, shown in Fig. 4. All these datasets consist of bi-temporal images captured by two distinct sensors depicting the change events of the lake, construction, and so on. Each pair of bi-temporal images spans a different geographical area and varies in size (width or height) from 300 to 2000 pixels and in resolution from 0.52 to 30 m. We chose these datasets to evaluate the comparison methods' performance adequately. Further detailed description of these datasets is given in Table II.

2) Evaluation Indicators: We selected two types of evaluation indicators for experiments to evaluate the performance of various methods quantitatively and comprehensively. One type is the quality evaluation indicator for CIMs, specifically the area under the receiver operating characteristic curve (AUR) and the area under the precision–recall (PR) curve (AUP). AUR and AUP have value ranges between 0 and 1, with values closer to 1 indicating better DI separability and higher quality. Therefore, we utilize AUR and AUP in the experiment to quantitatively compare and analyze the CIMs generated by different methods. In addition, the other category includes evaluation metrics for BCDIs. In this study, we selected three commonly used comprehensive evaluation indicators in the

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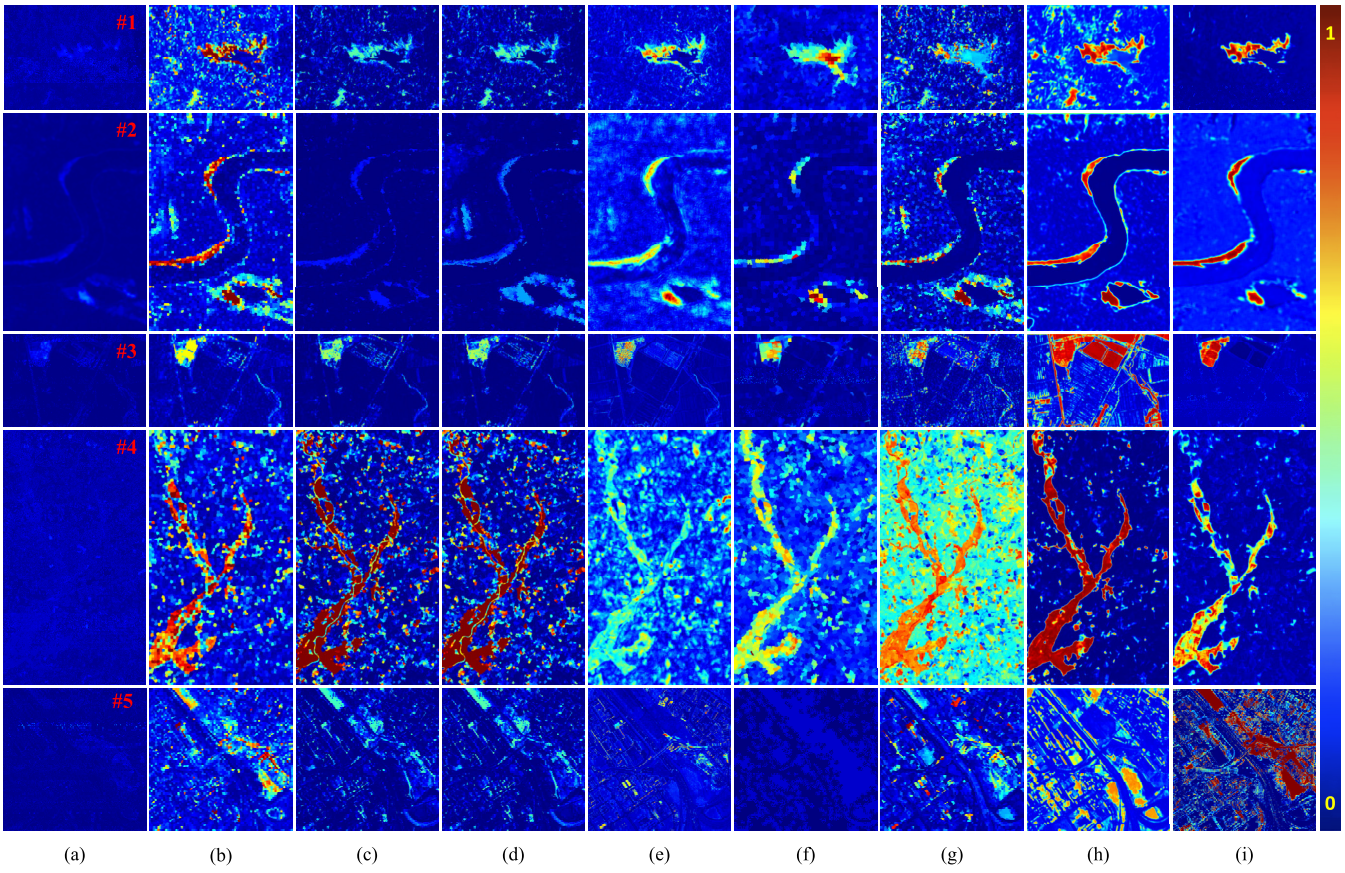


Fig. 5. CIMs of different methods on datasets #1–#5: (a) INLPG [46], (b) IRG-McS [47], (c) SCASC [48], (d) GIR-MRF [45], (e) CACD [35], (f) SR-GCAE [32], (g) SDC-GAE [36], (h) PRBCD [52], and (i) ours.

field of the CD in [35], [45], and [50]: overall accuracy (OA), kappa coefficient (KC), and $F1$-score ($F1$) to numerically compare and analyze the BCDIs. In the ablation analysis part, we also introduced the mean intersection over union (mIOU) to measure the performance of different model components.

B. Comparative Methods and Experimental Setup

1) Comparative Methods: To demonstrate the superiority of the proposed CFAT-MCD, we chose nine benchmark methods for comparison in this experiment. Concretely, there are four well-established traditional methods: INLPG [46], IRG-McS [47], SCASC [48], and GIR-MRF [45], and seven classic DL-based methods: SCCN [51], X-Net [33], ACE-Net [33], CACD [35], SR-GCAE [32], SDC-GAE [36], and PRBCD [52]. These methods are selected for comparative analysis due to their demonstrated strong performance across multiple MCD datasets. By comparing them with our proposed approach, we aim to validate the superiority and effectiveness of our approach.

2) Experimental Configuration: To ensure fairness in comparison, we follow the parameter settings outlined in the original articles of each method. For parameters without detailed specifications, we use trial and error to determine the optimal settings. The final performance of our proposed method is derived based on the pseudo-labels generated by the IRG-McS [47] approach. During the erosion operation, a kernel size of 7×7 is employed. Our method involves three hyperparameters: the size of patches (S_p), the number of pseudo-samples (N_{ps}), and the size of the window (S_w) in the SFCR. Parameters N_{ps} and S_w are consistently set to 100 and 3×3, respectively, across all datasets, while the value of S_p varies. Specifically, S_p is set to 11 for Dataset #4 and to 13 for the remaining datasets. A detailed analysis of the impact of these hyperparameters is provided in Section V. The proposed CFAT-MCD model is implemented using the PyTorch framework and trained on a single NVIDIA RTX 3090 GPU. Model parameters are optimized using the Adam optimizer with a learning rate of 0.0001. The number of training epochs and the batch size are set to 50 and 128, respectively. All experimental results presented in this section are obtained under the aforementioned settings.

C. Comparisons on CIMs

In this section, we comprehensively compare and analyze the quality of CIMs generated by different methods. Here, we selected methods that can obtain CIMs for comparison, including INLPG [46], IRG-McS [47], SCASC [48], GIR-MRF [45], CACD [35], SR-GCAE [32], SDC-GAE [36], and PRBCD [52]. Specifically, we first compare the visualization results of CIMs. Then, we use the ROC and PR curves and the corresponding areas under the curves AUR and AUC to analyze the quality of CIMs of different methods.

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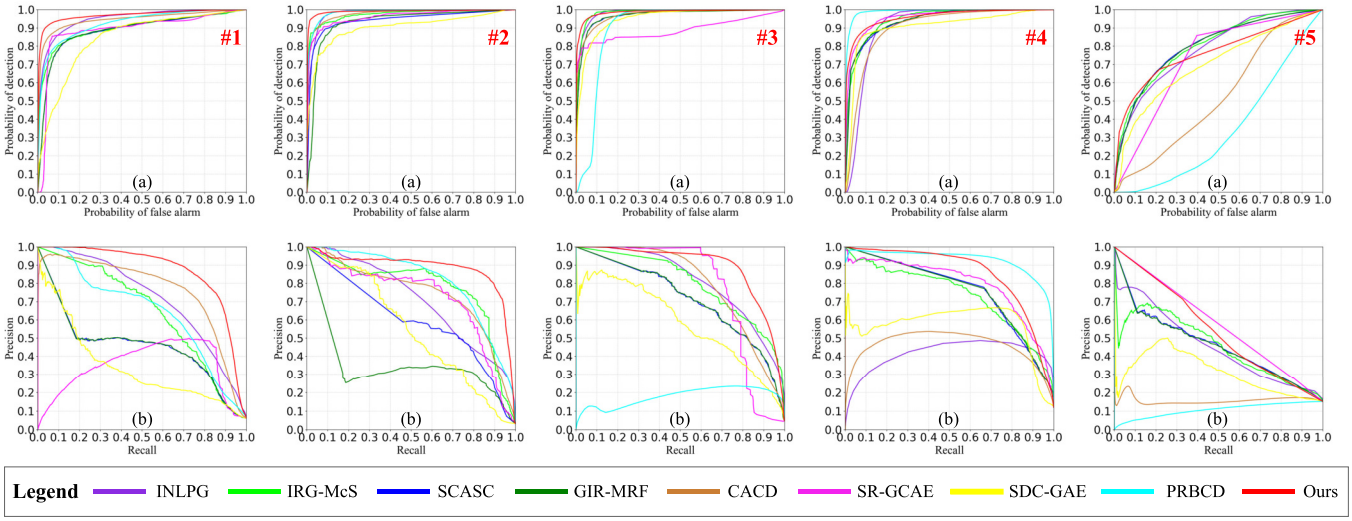


Fig. 6. Confidence comparisons with different methods on datasets #1–#5: (a) ROC curves and (b) PR curves.

TABLE III

COMPARISONS BETWEEN THE PROPOSED METHOD AND RELATED METHODS ON AUR AND AUP $\in [0, 1]$ FOR DATASETS #1–#5. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Dataset #1		Dataset #2		Dataset #3		Dataset #4		Dataset #5		Average	
	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP
INLPG [46]	0.950	0.721	0.981	0.707	0.984	0.811	0.912	0.413	0.786	0.456	0.923	0.622
IRG-McS [47]	0.899	0.643	0.973	0.773	0.984	0.798	0.946	0.735	0.799	0.454	0.920	0.680
SCASC [48]	0.886	0.456	0.942	0.594	0.967	0.710	0.938	0.758	**0.800**	0.463	0.907	0.596
GIR-MRF [45]	0.888	0.457	0.932	0.348	0.967	0.712	0.939	0.759	0.798	0.459	0.905	0.547
CACD [35]	0.947	0.763	0.978	0.742	0.971	0.796	0.902	0.459	0.524	0.158	0.864	0.583
SR-GCAE [32]	0.888	0.339	0.961	0.719	0.888	0.777	0.957	0.795	0.731	**0.578**	0.885	0.642
SDC-GAE [36]	0.844	0.341	0.903	0.544	0.951	0.581	0.906	0.571	0.700	0.309	0.861	0.469
PRBCD [52]	0.926	0.645	0.986	0.814	0.899	0.178	**0.990**	**0.923**	0.342	0.106	0.829	0.533
Ours	**0.971**	**0.871**	**0.990**	**0.882**	**0.986**	**0.891**	0.959	0.854	0.764	0.542	**0.934**	**0.808**

The CIMs of various methods on datasets #1–#5 are presented in Fig. 5. We can observe that, compared to other methods, our CIM can significantly highlight the changed areas. In unchanged areas, our CIM also shows the least hotspot coverage. From the visualization results, we can see that most methods can recognize the change region on datasets #2 and #3, and the number of false positives is relatively low. However, in the remaining three datasets, there are significant differences in the performance of different methods. For example, INLPG [46] performs well on datasets #1 and #5, but it is the worst of all methods on dataset #4. CACD [35] can accurately identify changes in datasets #1–#3 but misidentifies the size in datasets #4 and #5. Overall, the CIM generated by our proposed method is of the highest quality.

We also obtained the ROC and PR curves corresponding to the CIM of each method to demonstrate the separability of CIMs intuitively, as shown in Fig. 6. We can observe that the ROC and PR curves of our proposed CFAT-MCD can outperform other compared methods on most datasets. For instance, the ROC curve of our proposed method is closer to the upper left corner, which represents the high discriminative ability. The PR curve of our method is generally higher than other curves, indicating that the model has high accuracy in identifying changes. For datasets #4 and #5, although the proposed CFAT-MCD does not show a significant advantage on the ROC and PR curves, it still demonstrated suboptimal performance. Overall, the CIM generated by our model has a stronger discriminative ability, which is quite ideal for MCD tasks. In addition to observing the curves, we compared the performance of different methods using two quantitative indicators: AUR and AUP. The comparison result on datasets #1–#5 is shown in Table III. The average quantitative results in the table can directly demonstrate the superiority of our proposed method in terms of overall performance.

D. Comparisons on BCDIs

In this section, we further compare the BCDI results of all benchmark methods with the proposed CFAT-MCD to verify the superiority of our method. The BCDIs of each approach

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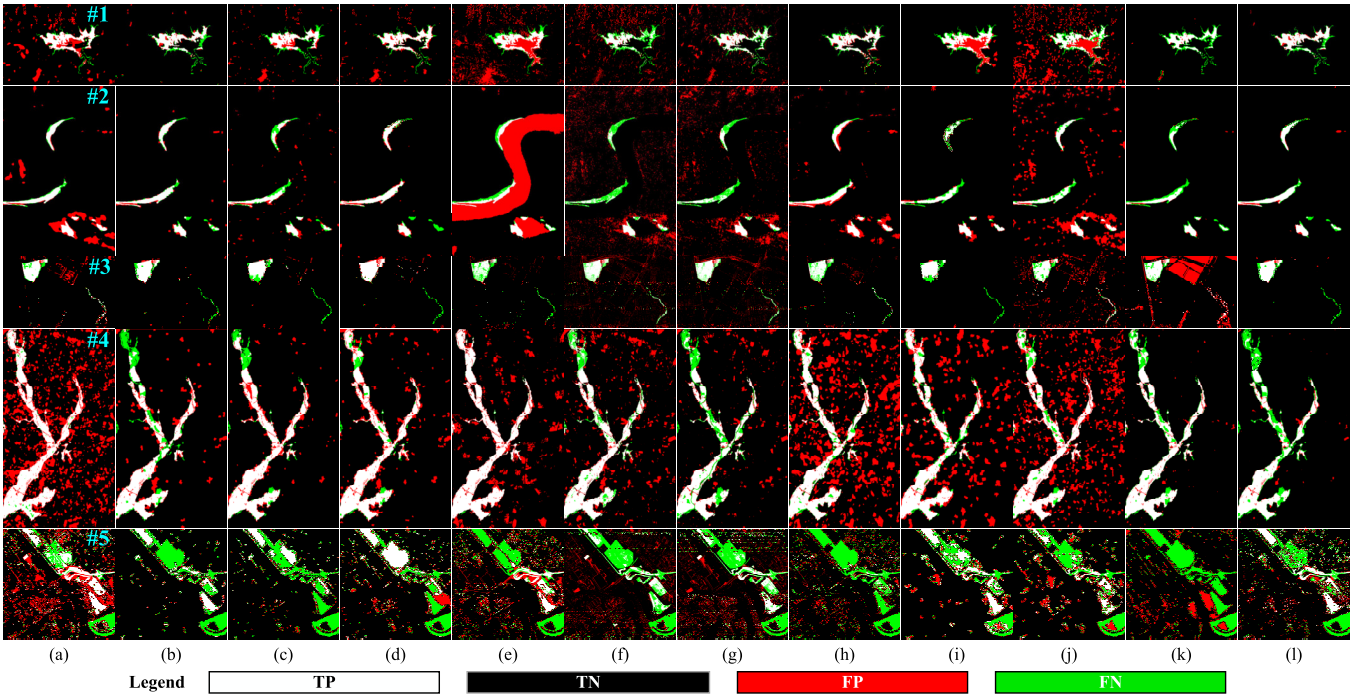


Fig. 7. BCDIs of different methods on datasets #1–#5: (a) INLPG [46], (b) IRG-McS [47], (c) SCASC [48], (d) GIR-MRF [45], (e) SCCN [51], (f) X-Net [33], (g) ACE-Net [33], (h) CACD [35], (i) SR-GCAE [32], (j) SDC-GAE [36], (k) PRBCD [52], and (l) ours. The BCDIs are colored for better observation.

TABLE IV

QUANTITATIVE COMPARISONS BETWEEN THE PROPOSED METHOD AND EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATASETS #1–#5. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Dataset #1			Dataset #2			Dataset #3			Dataset #4			Dataset #5			Average		
	OA	KC	F1	OA	KC	F1	OA	KC	F1	OA	KC	F1	OA	KC	F1	OA	KC	F1
INLPG [46]	0.918	0.519	0.558	0.955	0.529	0.550	0.968	0.683	0.700	0.794	0.432	0.531	0.796	0.335	0.455	0.886	0.500	0.559
IRG-McS [47]	0.971	0.739	0.754	0.986	0.788	0.795	0.983	0.794	0.803	0.936	0.704	0.740	**0.875**	0.349	0.403	0.950	0.675	0.699
SCASC [48]	0.947	0.593	0.621	0.976	0.623	0.636	0.979	0.741	0.751	0.950	**0.776**	**0.804**	0.872	0.327	0.381	0.945	0.612	0.639
GIR-MRF [45]	0.957	0.674	0.697	0.986	0.788	0.795	0.981	0.798	0.807	0.937	0.734	0.770	0.873	0.447	0.519	0.947	0.688	0.718
SCCN [51]	0.862	0.336	0.395	0.792	0.127	0.177	0.975	0.672	0.685	0.922	0.708	0.751	0.820	0.224	0.326	0.874	0.413	0.467
X-Net [33]	0.918	0.340	0.443	0.918	0.232	0.268	0.954	0.595	0.618	0.909	0.637	0.688	0.830	0.229	0.322	0.906	0.407	0.468
ACE-Net [33]	0.935	0.549	0.582	0.928	0.297	0.329	0.956	0.613	0.635	0.928	0.659	0.701	0.842	0.294	0.382	0.918	0.482	0.526
CACD [35]	0.975	0.776	0.790	0.967	0.614	0.630	0.978	0.738	0.750	0.798	0.417	0.516	0.829	0.029	0.085	0.909	0.515	0.554
SR-GCAE [32]	0.937	0.546	0.579	0.981	0.694	0.704	0.982	0.761	0.770	0.885	0.586	0.649	0.874	**0.460**	**0.531**	0.932	0.609	0.647
SDC-GAE [36]	0.833	0.259	0.328	0.915	0.343	0.375	0.935	0.490	0.521	0.836	0.447	0.522	0.834	0.266	0.358	0.870	0.361	0.421
PRBCD [52]	0.969	0.674	0.689	0.983	0.663	0.671	0.865	0.315	0.365	0.841	0.489	0.572	0.790	-0.091	0.002	0.889	0.410	0.460
Ours	**0.976**	**0.779**	**0.791**	**0.990**	**0.833**	**0.838**	**0.985**	**0.820**	**0.828**	**0.951**	0.742	0.769	0.870	0.398	0.467	**0.955**	**0.714**	**0.739**

are displayed in Fig. 7 and the quantitative result is listed in Table IV. From the visualization results, it can be seen that the BCDI generated by our method is more similar to the ground-truth map, with more complete change areas and a cleaner background. The coloring results also show that the number of FPs and FNs is lower than other comparison methods. For example, as shown in Fig. 7(a), (e), (h), and (i), although these methods can effectively detect the changed area, there are a large number of false positive pixels in the results. Based on the results from different datasets, datasets #1–#3 showed good overall detection performance. In contrast, datasets #4 and #5 showed a large number of FPs and TNs, respectively. It can thus be seen that the detection difficulty of datasets #4 and #5 is higher than that of other datasets. All comparison methods, including our model, performed poorly on dataset #5, primarily because this dataset has the largest image size but the lowest resolution, resulting in significant noise and sample imbalance issues. Overall, IRS-McS [47], SCASC [48], GIR-MRF [45], ACE-Net [33], PRNCD [52], and our method demonstrated excellent performance in the visualization results.

Additionally, the quantitative results in Table IV provide a detailed comparison of the performance differences between various methods. We have bolded the best results in each group of experiments. It can be seen that our CFAT-MCD shows comprehensive performance superiority in datasets #1–#3. In datasets #4 and #5, our method ranks third in performance and is very close to the second-best method. Finally, our

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TABLE V
ABLATION AND COMPARISON RESULTS OF THE INITIAL DETECTION AND INITIAL DETECTION WITH OUR CFAT-MCD ON DATASETS #1–#5

Methods	Dataset #1				Dataset #2				Dataset #3				Dataset #4				Dataset #5			
	Initial		with Ours		Initial		with Ours		Initial		with Ours		Initial		with Ours		Initial		with Ours	
	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1	KC	F1
INLPG [46]	0.519	0.558	**0.686**	**0.708**	0.529	0.550	**0.726**	**0.735**	0.683	0.700	**0.793**	**0.801**	0.432	0.531	**0.773**	**0.803**	0.335	0.455	**0.453**	**0.545**
IRG-McS [47]	0.739	0.754	**0.779**	**0.791**	0.788	0.795	**0.833**	**0.838**	0.794	0.803	**0.820**	**0.828**	0.704	0.740	**0.742**	**0.769**	0.349	0.403	**0.387**	**0.467**
SCASC [48]	0.593	0.621	**0.683**	**0.705**	0.623	0.636	**0.817**	**0.823**	0.741	0.751	**0.779**	**0.787**	0.776	0.804	**0.819**	**0.840**	0.327	0.381	**0.453**	**0.509**
GIR-MRF [45]	0.674	0.697	**0.742**	**0.758**	0.788	0.795	**0.826**	**0.832**	0.798	0.807	**0.852**	**0.858**	0.734	0.770	**0.860**	**0.877**	0.447	0.519	**0.485**	**0.553**
SCCN [51]	0.336	0.395	**0.545**	**0.573**	**0.127**	**0.177**	0.092	0.143	0.672	0.685	**0.773**	**0.782**	0.708	0.751	**0.845**	**0.865**	0.224	0.326	**0.279**	**0.358**
X-Net [33]	0.340	0.443	**0.642**	**0.658**	0.232	0.268	**0.317**	**0.326**	0.595	0.618	**0.774**	**0.783**	0.637	0.688	**0.729**	**0.757**	0.229	0.322	**0.294**	**0.343**
ACE-Net [33]	0.549	0.582	**0.715**	**0.729**	0.297	0.329	**0.718**	**0.728**	0.613	0.635	**0.720**	**0.730**	0.659	**0.701**	**0.665**	0.695	0.294	0.382	**0.339**	**0.386**
CACD [35]	0.776	0.790	**0.783**	**0.795**	0.614	0.630	**0.780**	**0.788**	0.738	0.750	**0.751**	**0.760**	0.417	0.516	**0.605**	**0.660**	0.029	0.085	**0.062**	**0.093**
SR-GCAE [32]	0.546	0.579	**0.552**	**0.583**	0.694	0.704	**0.756**	**0.765**	0.761	0.770	**0.787**	**0.795**	0.586	0.649	**0.805**	**0.830**	0.460	0.531	**0.521**	**0.596**
SDCGAE [36]	0.259	0.328	**0.608**	**0.635**	0.343	0.375	**0.431**	**0.455**	0.490	0.521	**0.816**	**0.824**	0.447	0.522	**0.689**	**0.724**	0.266	0.358	**0.313**	**0.414**
PRBCD [52]	0.674	0.689	**0.695**	**0.709**	0.663	0.671	**0.772**	**0.779**	0.315	0.365	**0.379**	**0.422**	0.489	0.572	**0.780**	**0.803**	**-0.091**	**0.002**	-0.125	0.001

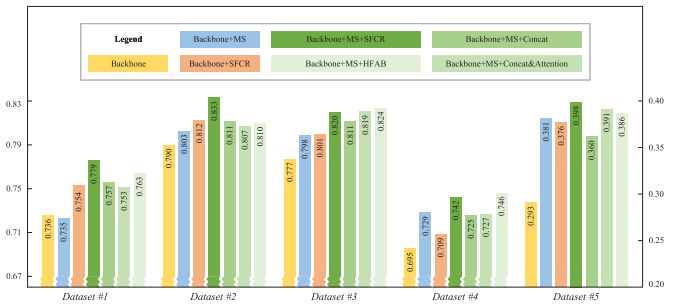


Fig. 8. Ablation and comparison results of our CFAT-MCD with different components on datasets #1–#5.

method also achieved the best performance in terms of average performance across the five datasets. Compared to the second-best GIR-MRF [45], we lead by 0.053 in the KC metric and 0.021 in the $F1$-score. Since the OA metric results are already very high, the gap between us and the second-best method is not significant, with a lead of only 0.005. Through the above comparison of visualization and quantification results, the superiority of our CFAT-MCD method has been fully verified.

V. DISCUSSIONS

In this section, we will first verify and analyze the effectiveness of different submodules in the proposed method. Then, the two parameters involved in the proposed method will be discussed and analyzed. The details are as follows.

A. Ablation Analysis

1) Effectiveness and Robustness of CFAT-MCD: To fully validate the effectiveness and robustness of our CFAT-MCD, we designed ablation experiments for CFAT-MCD and comparison experiments for different quality pseudo-labels. We utilize the eleven comparison methods introduced in Section IV-B to generate initial detection results and train our CFAT-MCD model based on these results. We present the initial detection results and the CFAT-MCD detection results based on the initial results in Table V. From the test results in the table, we can see that our method can further improve performance based on almost all initial results. Among them, the model based on the ACE-Net [33] achieved the maximum improvement, increasing K1 by 0.421 and $F1$-score by 0.399 on dataset #2, except for a few cases, such as the results of SCCN [51] on dataset #2 and PRBCD [52] on dataset #5. Due to the extremely low performance of the initial detection results, the pseudo-labels became completely unreliable, misleading the learning of our network and further degrading the detection performance. First, in most cases, the introduction of our CFAT-MCD can improve detection performance, providing sufficient proof of the effectiveness of our method. Second, except in extreme cases where the quality of pseudo-labels is completely unreliable, our CFAT-MCD can still improve detection performance, even when the $F1$-score is as low as 0.322. This result sufficiently demonstrates that our method exhibits good robustness in the presence of data noise and unreliable samples.

2) Effectiveness of CFAT Components: To test the effect of each component, we decompose the proposed CFAT into three parts: the backbone Transformer network, MS fusion, and the SFCR module. Based on this, we conducted ablation studies on the effect of different module combinations on performance. The quantitative results of the ablation experiment are presented in Table VI. We have selected the KC indicator results for visualization and plotted them in Fig. 8. The data in Table VI shows that the introduction of MS and SFCR can effectively improve the performance of the backbone network. Fig. 8 provides a more intuitive understanding of the contributions of different components. We can observe that introducing MS or SFCR individually yields different improvements across different datasets. However, introducing MS and SFCR simultaneously achieves significant performance improvements across all experimental data. The above findings sufficiently demonstrate the effectiveness of our model components.

3) Effectiveness of SFCR: Our SFCR is a novel spatial–frequency feature fusion module. In single-modal CD studies, numerous studies have successfully integrated spatial and frequency features. However, in MCD tasks, there are currently no spatial–frequency feature strategies for multimodal images. Therefore, to verify the effectiveness of our SFCR fusion strategy, we introduced three other different fusion strategies for comparison. Two of these strategies are simple feature concatenation and feature concatenation with channel

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TABLE VI
QUANTITATIVE RESULT OF ABLATION AND COMPARISON EXPERIMENTS FOR DIFFERENT COMPONENTS ON DATASETS #1–#5

Conponents	Dataset #1			Dataset #2			Dataset #3			Dataset #4			Dataset #5		
	KC	mIOU	F1	KC	mIOU	F1	KC	mIOU	F1	KC	mIOU	F1	KC	mIOU	F1
Backbone	0.736	0.782	0.765	0.790	0.822	0.801	0.777	0.813	0.789	0.695	0.745	0.735	0.293	0.570	0.431
Backbone+MS	0.735	0.784	0.761	0.803	0.834	0.809	0.798	0.829	0.810	0.729	0.777	0.759	0.381	0.576	0.449
Backbone+SFCR	0.754	0.794	0.783	0.812	0.835	0.817	0.801	0.811	0.813	0.709	0.765	0.740	0.376	0.573	0.445
Backbone+MS+SFCR	**0.779**	**0.815**	**0.791**	**0.833**	**0.855**	**0.838**	**0.820**	**0.846**	**0.828**	0.742	0.786	0.769	**0.398**	**0.584**	**0.467**
Backbone+MS+Concat	0.757	0.800	0.770	0.811	0.839	0.817	0.807	0.836	0.815	0.725	0.774	0.753	0.360	0.564	0.442
Backbone+MS+CA [53]	0.753	0.797	0.766	0.819	0.845	0.825	0.812	0.840	0.820	0.727	0.775	0.759	0.391	0.580	0.458
Backbone+MS+HFAB [26]	0.763	0.804	0.775	0.824	0.849	0.830	0.810	0.838	0.817	**0.746**	**0.788**	**0.775**	0.386	0.578	0.445

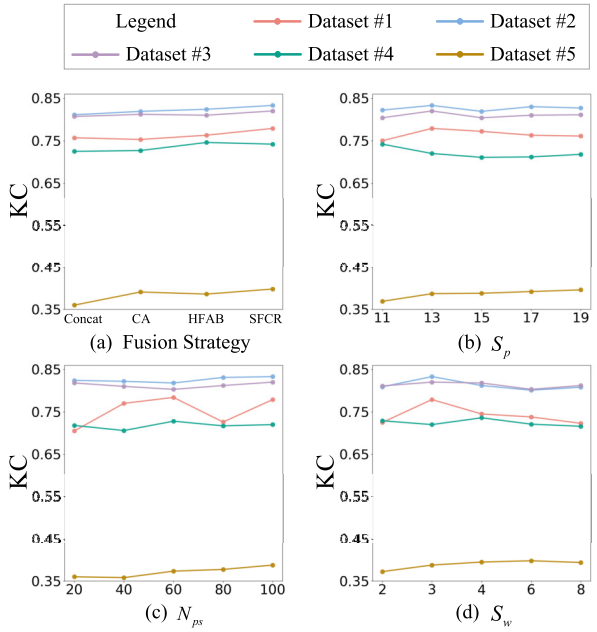


Fig. 9. Variables analysis of detection performance for our CFAT-MCD on datasets #1–#5. (a) Fusion strategy, (b) patch size, (c) number of pseudo-samples, and (d) window size of SFCR.

attention mechanisms. The other fusion strategy is a high-frequency attention block (HFAB), which comes from the classic method of utilizing frequency information in building CD, HFA-Net [2]. We compared the HFAB designed for frequency information in this network with our SFCR. The results of the comparative experiments for these three fusion strategies are shown in the lower part of Table VI, and we have visualized the experimental results of the KC indicator, as shown in Fig. 9(a). The experimental results show that, except for dataset #4, HFAB outperforms our method. On the remaining datasets, our SFCR fusion strategy demonstrates a significant performance improvement. This result sufficiently proves the effectiveness of SFCR in spatial–frequency feature fusion.

B. Parameter Analysis

In our approach, the settings of the patch size, the number of pseudo-samples, and the window size have a significant impact on model performance. Therefore, it is necessary to explore and discuss the settings of these parameters for different data scenarios.

1) Patch Size (S_p): This section delves into the effect of varying S_p on detection performance across datasets #1–#5. Utilizing the control variable method, we fix N_{ps} at 100 and (S_w) at 3 for all five datasets, while the range for S_p is systematically adjusted between [11, 13, 15, 17, 19]. Based on these settings, we plot a line graph illustrating the variations in accuracy (in KC) across different values of S_p for datasets #1–#5, as depicted in Fig. 9(b). It is worth noting that S_p causes the least fluctuation in model performance among the three hyperparameters. However, as S_p increases, the change in model performance varies across different datasets. For example, datasets #1–#3 achieve optimal performance when S_p is 13. However, dataset #4 achieves optimal performance when S_p is 11. As S_p increases, model performance begins to decline. Dataset #5 is entirely different, with its detection performance increasing as S_p increases. This may be because the image size of dataset #5 is larger, and larger patches can capture richer and varied information.

2) Number of Pseudo-Samples (N_{Ps}): In addition to the parameter S_p, the parameter N_{ps} may also influence the detection performance of MCD. To evaluate its impact on model accuracy, comprehensively analyze N_{ps} by varying its value from 20 to 100 while keeping S_p fixed at 13 and S_w fixed at 3 across all datasets. This setting allows us to examine the effect of N_{ps} on detection performance. Fig. 9(c) illustrates the accuracy (in KC) variation curves of different N_{ps} on datasets #1–#5. As can be seen from the figure, the peak of dataset #4 occurs when N_{ps} is 60. The peaks of the other four datasets all occur when N_{ps} is 100. This may be due to the improved confidence of the samples after corrosion and balancing. Therefore, using more pseudo-label samples can improve the model training effect to a certain extent, thereby leading to an increase in detection performance.

3) Window Size (S_w) of SFCR: S_w directly affects the calculation range of the window-based DCT in SFCR. The motivation behind window design is to focus on local frequency characteristics. By analyzing the frequency components within windows at different positions, it is possible to capture the spatial structural relationships of images better. To verify the effect of S_w, we selected five values, 2, 3, 4, 6, and 8, while fixing S_p at 13 and N_{ps} at 100. The experimental results are shown in Fig. 9(d). The experimental results show that the performance peaks of datasets #1–#3 occur when S_w is set to 3. Datasets #4 and #5 perform best when S_w is set to 4 and 6, respectively. It is worth

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noting that when the value of S_w is less than 3 or greater than 6, the model's detection performance shows a significant decline across all five datasets. This indicates that if S_w is too small, the extracted spectrum resolution will be extremely low, making it difficult to obtain practical information. If S_w is too large, it will easily mask local feature changes and lose spatial position information. Therefore, the optimal setting for S_w needs to find a balance between local frequency-domain features and spatial position information.

C. Computational Complexity

Finally, we conducted a detailed analysis of the computational complexity of each stage in our proposed method. As mentioned in Section III, our CFAT-MCD consists of three main stages: pseudo-label generation, sample generation, and detection. Among them, the detection stage dominates the computational cost. The detection stage includes our CFAT and BCDI generation operations. Taking dataset #1 as an example, CFAT's inference time reaches 21.35 s with total FLOPs of 65.51 T and 8.69 M parameters, which is significantly higher than any other operations. The total runtime of our CFAT-MCD exceeded 25 s. Our method achieves better detection performance at the cost of increased computational complexity. This increase mainly results from our patch-wise image-processing strategy and the introduction of an MS Transformer-based architecture. While these designs significantly enhance detection accuracy, they also lead to increased inference costs. In future work, we aim to develop more efficient architectures to reduce computational overhead without compromising performance.

VI. Conclusion and Future Works

In this article, we propose an unsupervised framework to address the task of MCD in the unlabeled RSI. CFAT-MCD is a hybrid framework that combines traditional and DL-based methods, which guides the learning direction of CFAT through the generated pseudo-labels. With a few balanced pseudo-samples, our CFAT-MCD can achieve more accurate recognition of land-cover changes. Besides, to enhance the utilization of the self-attention mechanism and mitigate the adverse effects caused by modal differences, we design the MS perception structure of CFAT and efficiently map the frequency features into the spatial domain through the proposed SFCR module. By combining the ideas of multimodal, MS, and multidomain, we enhance the robustness and discriminative properties of the extracted features, which helps the model capture the change patterns better. CFAT can be regarded as a powerful postprocessing module capable of generating more fine-grained change maps based on existing methods. Comparative experiments and quantitative analysis on four MCD datasets validate the effectiveness of the proposed method in this article. Despite the strong performance of our proposed method in terms of detection accuracy and generalization ability, it still exhibits several limitations that warrant further exploration and refinement. First, CFAT-MCD is a multistage detection framework based on Transformer architectures. The overall computational complexity remains high, primarily due to the patch-wise processing strategy and the MS spatial–frequency feature fusion employed in the detection network. While these computations are essential for addressing the challenging issue of modality discrepancy, they pose obstacles to efficiency and scalability. Second, as an unsupervised method relying on pseudo-labels, the performance of CFAT-MCD is inherently dependent on the quality of the pseudo-labels generated in the early stage. When these pseudo-labels are unreliable or contain only few true information, the model struggles to learn effective representations, thereby limiting its detection capability.

Inspired by the above limitations, several directions can be considered to improve and extend our method in future work. On the one hand, we can start by lightweighting the network structure. By exploring new types of efficient basic operators, higher detection performance can be achieved at a lower computational cost. Additionally, model compression techniques, such as knowledge distillation or pruning, may help eliminate redundant computations without compromising performance. On the other hand, since pseudo-label quality fundamentally constrains model performance, future work could try to design an iterative self-refinement mechanism. This mechanism can guide and improve the reliability of the pseudo-labels through the model's output or a few external prior knowledge inputs. Furthermore, multibranch consistency learning can be introduced to filter and fuse pseudo-labels generated from different perspectives, thereby enhancing the overall quality of these labels. Finally, in the era of foundation models, the ability to efficiently process large-scale scene-level data has become a critical requirement for valuable models. To extend our CFAT-MCD to large-scale applications, one promising direction is to employ our framework as a proxy model. In such a setup, a powerful foundation model could first perform global perception over large-scale data while the proxy model ensures alignment with the specific target task, enabling effective and scalable CD.

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